Contemporary Problem Solving

Project Ideas

DevOps

1. **Docker containers**. Dockerize a range of popular web application stacks (e.g. React–Express–MongoDB, or React-Spring-MySQL, etc. Simulate a real-world deployment to a server.
2. **CI/CD Pipeline.** Set up a CI/CD pipeline using appropraite tools, like Jenkins, to automate the build, test, and deployment processes.
3. **Container orchestration.** Deploy and manage containerized applications using Kubernetes.

Web Applications

1. **Infinite Scroll**. Create an interface which reads large amounts of data from an API and loads more as the user scrolls down or up.
2. **Uploading Images**. Build a front-end prototype to process image (type, size, crop) before uploading and a back end to verify and create thumbnails.
3. **Uploading files**. Upload any type of file, identify the type, sanitize, and store it accordingly.
4. **Doctor appointments**. Design and implement the database and a proof-of-concept web application.
5. **Restaurant booking**. Design and implement the database and a proof-of-concept web application.
6. **Flight, Bus, or Theater seat reservation**. Design and implement the database and a proof-of-concept web application using SVG UI.
7. **Online Payments**. Implement a prototype and use PayPal and Stripe sandboxes to demonstrate online payments how-to.
8. **Multilingual Web Apps**. Design and build a proof-of-concept web application displaying information in many languages, involving a database.
9. **Localize Web Apps**. Design and build a system where dates, times, currency, number format, weight, length, etc., will be displayed in the local format of the user.

10. **Front-end Authentication and Authorization**. Use a front-end framework like React and examine how JWT can be used with a simple back end.
11. **Flight Monitoring**. Use appropriate API to display the position of a flight requested by the user, on a map.
12. **Microservices**. Develop an example web application using microservices.
13. **Academic Journal**. Use a CMS (WP, Drupal) to build an academic journal web site which can be indexed in Google Scholar.

e-Learning

1. **Sharable Content Object Reference Model (SCORM)**. Build a prototype tool to help creating SCORM content.
2. **E-learning Course**. Find reliable open-source tools and create an interactive e-learning course.

Data Science - Machine Learning

1. **Image Pattern recognition**. Identify specific objects (coins, vehicles, letters, etc.) in any given image.
2. **Voice recognition**. Create a simple game (e.g. snake) and control using voice commands.
3. **Anonymize Data**. Find appropriate methods to anonymize data so that they can become publicly accessible.
4. **Chatbot**. Build a chatbot that can be used on Discord (and maybe adopted by the Coding Society group).
5. **Sentiment Analysis**. Retrieve crowd generated data (comments on e-shop, posts on social media, etc.) and characterize as positive, negative, or neutral.
6. **Web Scraping**. Collect data from websites and use them to analyze a topic.
7. **Generate realistic data**. Build an application to generate random yet realistic data for some problem, e.g. car sensors values.
8. **Solve a Kaggle ML project**.

Mobile Applications

1. **City Problems**. Build an app to monitor problems in the city and their coordinates and transmit to a server. Display problems on a web page.
2. **Shopping Basket Cost**. Read barcodes of products, let user type quantity, retrieve price from a server and display the total cost.

3. **OCR**. Using the camera, read the name of a product and then the price from the price tag, and transmit to a database, along with date.
4. **Extreme Sport Athlete Location Monitoring**. Provide the location of the athlete to a backend monitoring system. Produce alert in case of large amount time without location submission. Solve the problem of areas

NFC - RFID

1. **Entry Control System**. Use mobile phone or NFC/RFID card to control entrance to a room, building, etc. Record all activity to a server.
2. **Virtual Tour Guide**. Use NFC/RFID tags to provide info of museum exhibits, national garden, etc.

Internet of Things (IoT)

1. **Noise Recording**. Record the noise level in an area (several points), transmit to a server and provide info on a web page.
2. **Patient Vital Signs**. Monitor patient vital signs, transmit to a server and alert relatives or the doctor in case of emergency.
3. **Farm Monitoring**. Build a device to monitor temperature, humidity, soil moisture, rainfall, and other data. Transmit to a server for the farmer.

ETC

1. **College Library Gamification**. Build an application which will motivate students to use the college library.
2. **Create PDF documents**. Build an application to generate pdf documents from given data. For example, an invoice.

Appendix 1: Acronyms

CI/CS:	Continuous Integration / Continuous Deployment
CMS:	Content Management System

IoT:	Internet of Things
NFC:	Near-field communication
RFID:	Radio-frequency identification
WP:	Wordpress